

# AlphaFold for complexes, and how to judge them

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A short introduction before Part 2

Cilia Connect 2026 · AI Structure Prediction Workshop · E. Lorentzen

# From one chain to two

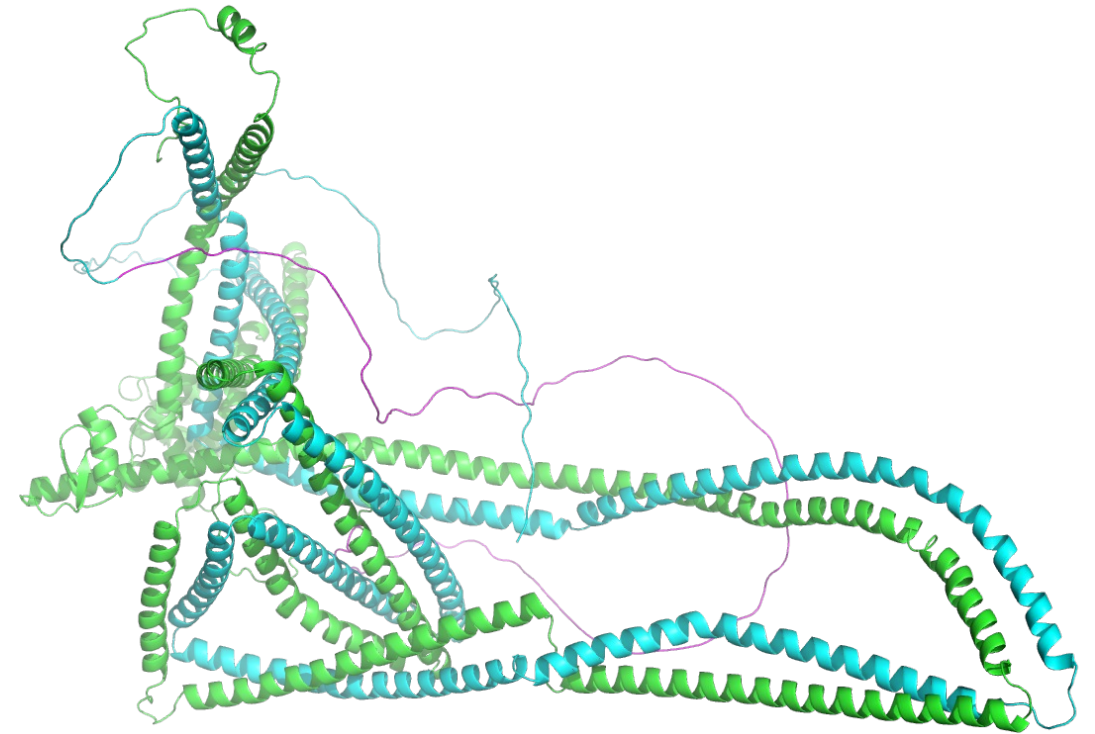
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- In Part 1 you asked of a single model: is this fold right?
- For two proteins the question changes: do they actually bind, and if so, how?
- Same server, same idea. But now we need a score for the interface between the chains, not just for each chain on its own.

# AlphaFold2 was trained on single proteins

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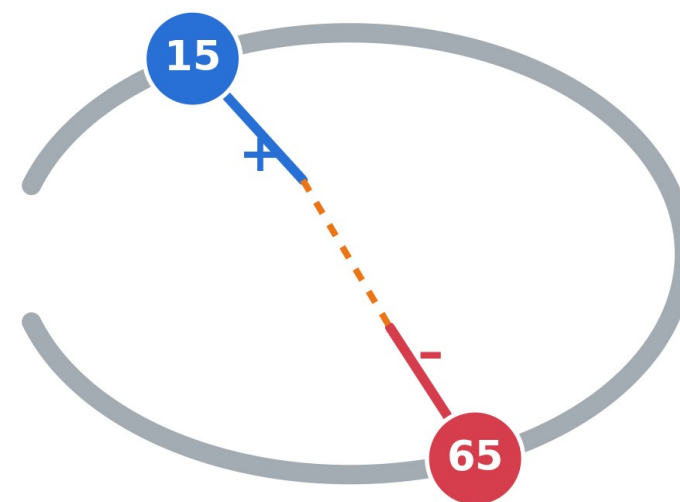
- AF2 (Jumper et al., Nature 2021) was trained on single protein chains, and fairly short ones.
- It took one sequence at a time: no complexes.
- Researchers soon found a trick: join two proteins with a poly-glycine linker into one sequence, and AF2 would often predict the complex faithfully.



The two proteins IFT81 and IFT74 (green, cyan) predicted as a complex, joined by a poly-glycine linker (magenta).

# Why does that work?

- An interface is physically much like a piece of protein core: the same hydrophobic packing, hydrogen bonds and salt bridges that hold one chain together.
- So a network that learned to fold single proteins already learned much of the physics of an interface.
- And interface residues co-evolve across species: a paired alignment of the two proteins carries the contact signal, just as it does within one protein.

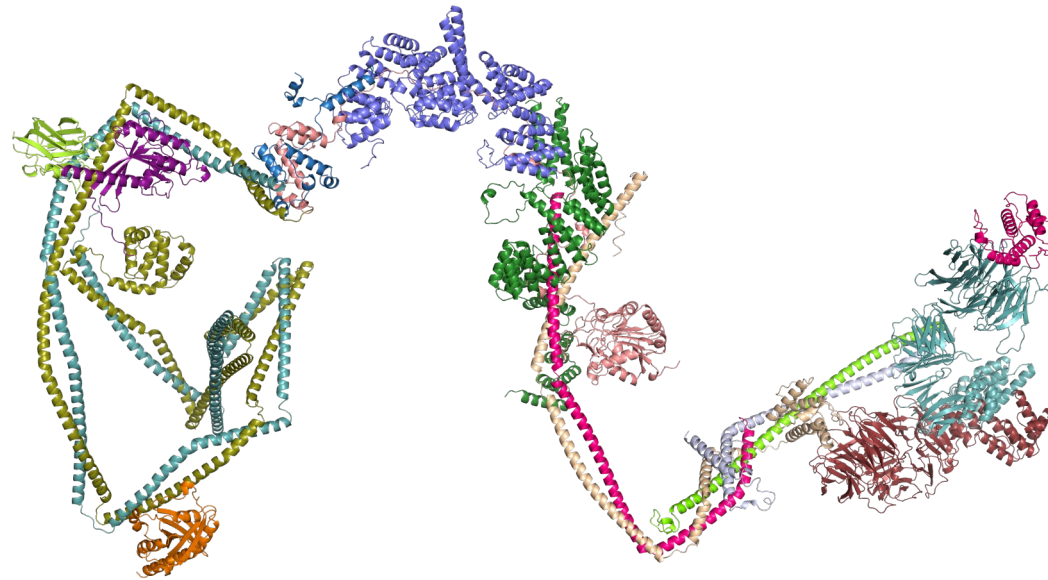


Co-evolving residues (here a salt bridge): correlated changes across species point to a 3D contact.

# AlphaFold-Multimer: trained on complexes

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- The linker was a clever workaround, not a real solution.
- AlphaFold-Multimer was then trained specifically on protein complexes.
- It predicts the separate chains directly, with no linker needed, and is markedly more accurate at interfaces.



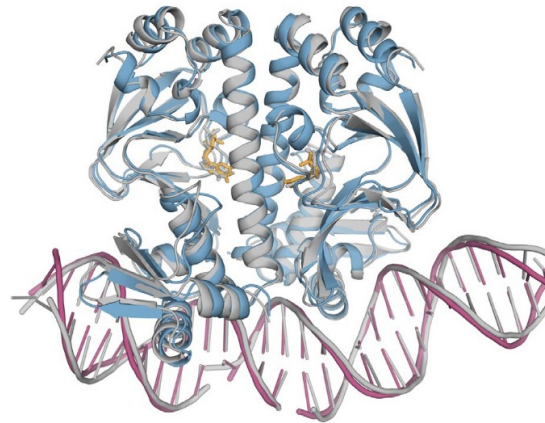
15 subunit IFT-B complex predicted by AF multimer

# AlphaFold3: one model for whole assemblies

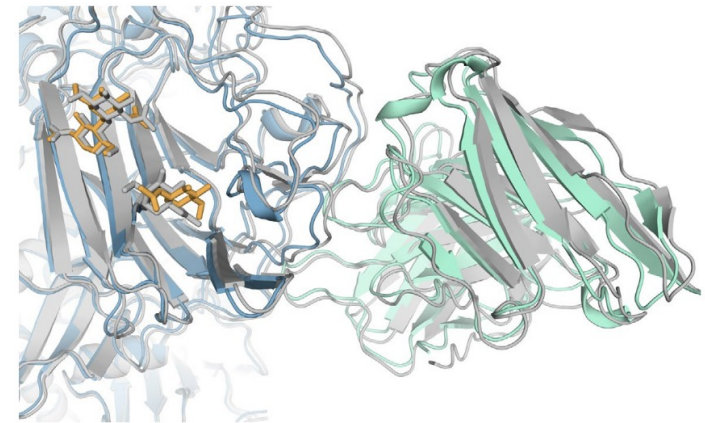
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- AF3 (Abramson et al., Nature 2024) replaces the older network with a diffusion model.
- It predicts several chains at once: proteins, DNA, RNA, ligands and ions, together in one job.
- This is the server you are using today.

**a**



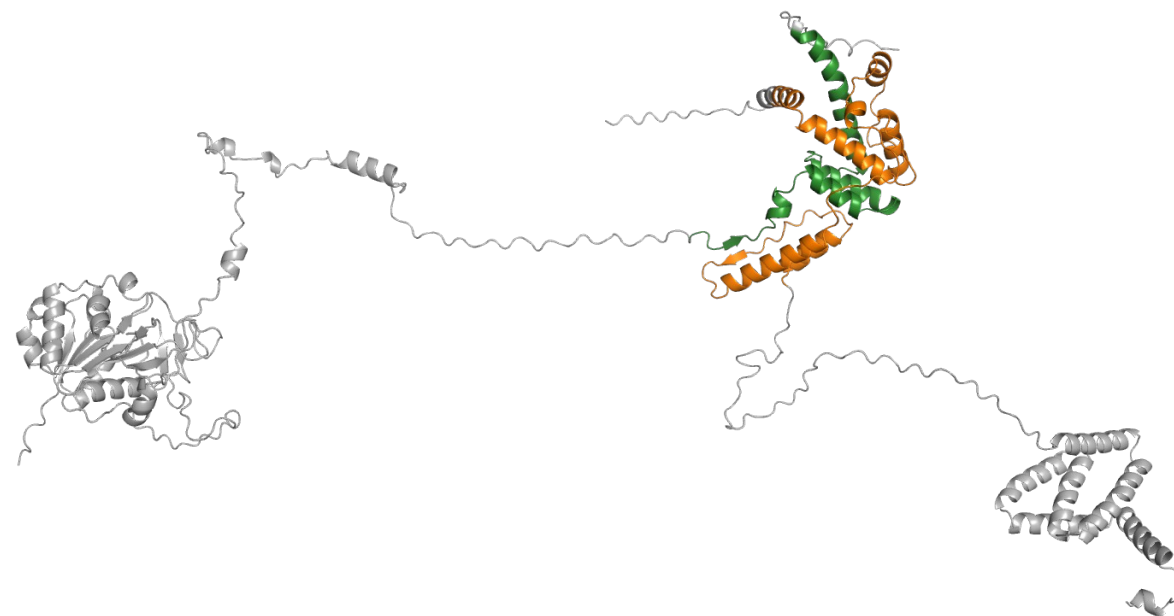
**b**



AF3 predicts large mixed assemblies in a single prediction.

# Scoring a complex: the ipTM

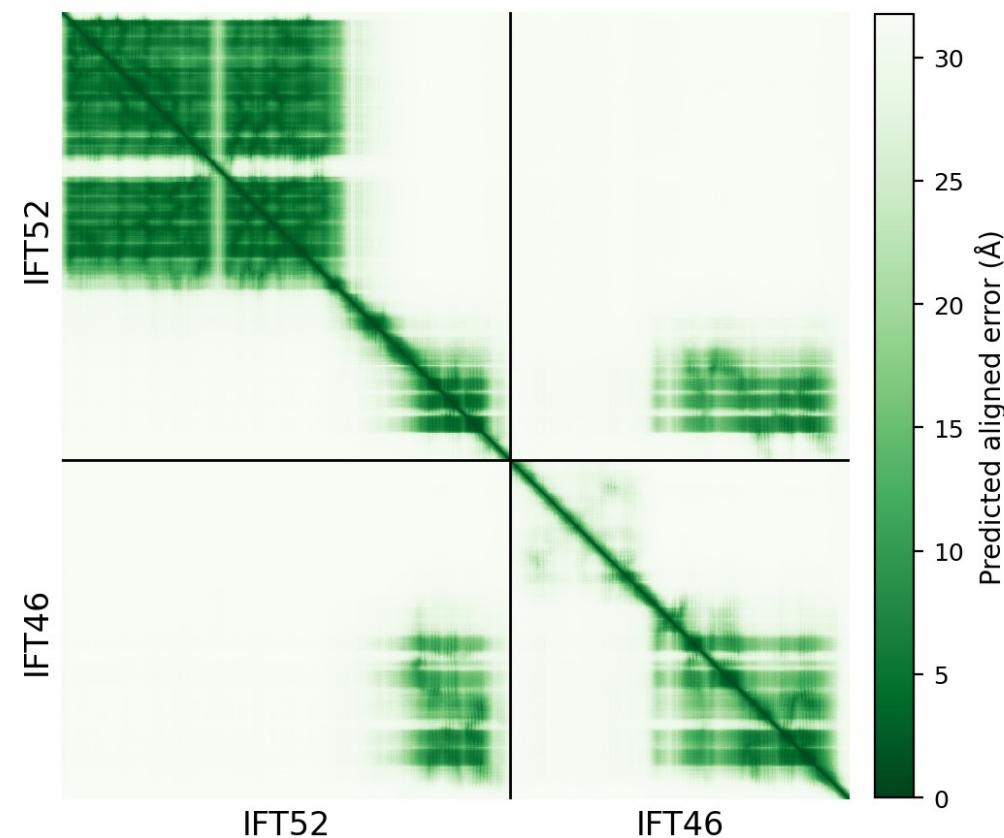
- ipTM (interface pTM) asks: how well is the arrangement between the chains predicted? One number, 0 to 1.
- Rough guide: above 0.8 is highly confident, 0.6 to 0.8 may be correct and below 0.6 may be wrong.
- The catch: it is a global average over all between-chain residue pairs. If two parts interact but the rest of each protein is disordered and floppy, those parts drag the average down.
- Ideally you would score only the interacting region, not the whole proteins.



A real, compact interface (green, orange) sits among large non-interacting parts (grey) that pull the ipTM down.

# The PAE shows if a complex is predicted with confidence and if so, where the interface is located

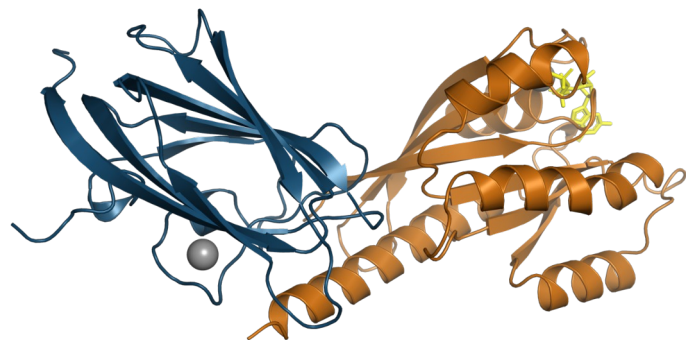
- The PAE gives, for a pair of residues, the expected error in their position relative to each other: dark = confident, light = not.
- For a complex, read the off-diagonal block between the two chains.
- A compact dark inter-chain block means the chains are confidently placed against each other: a real interface. A light block means they float apart.
- ipTM is really just the one-number summary of that block.



PAE of a two-chain prediction: look for a dark block off the diagonal, between the chains.

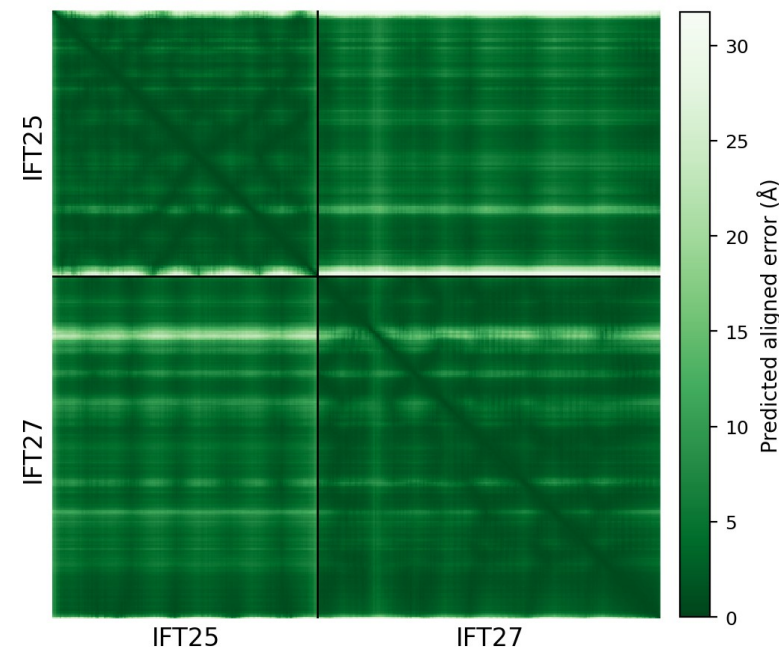
# A confidently predicted complex structure: IFT25-IFT27

IFT25 and IFT27 form a tight, constitutive heterodimer, and here the scores behave: a high ipTM and a PAE that is dark throughout, including between the chains. Compare the IFT52:IFT46 case you just saw (ipTM 0.35).



IFT25 (blue) and IFT27 (orange) interlocked; GTP bound in IFT27.

ipTM  
**0.91**  
highly confident (> 0.8)

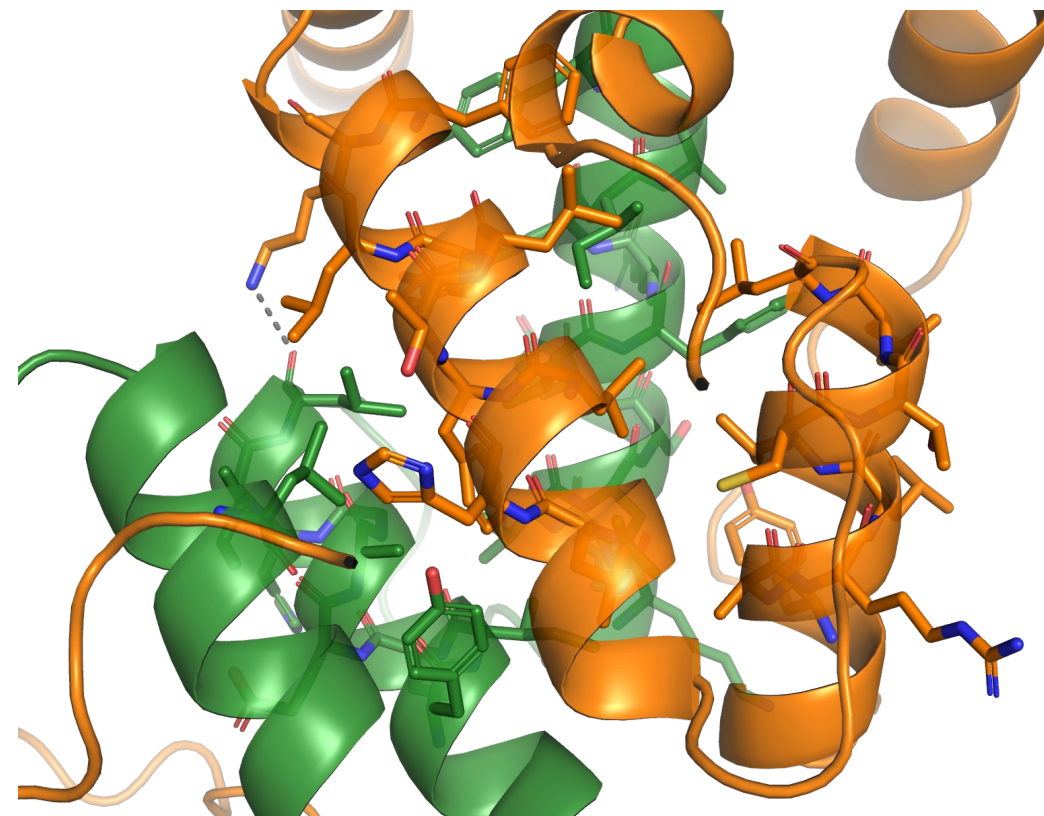


PAE dark across the whole plot: the chains are confidently placed.

## Next: your IFT52:IFT46 complex

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- You have submitted the full-length IFT52:IFT46 job.
- You will probably meet a low ipTM. Your task: decide whether that means no interaction, or a real interface hidden in a floppy pair of proteins.
- Read the PAE, find the interface for yourself, then test it by predicting a minimal complex.



What you are looking for: the confident core of a predicted interface.